Mathematics GCE O Level 2022 - Paper 1

Duration: 1h30 | Cameroon GCE Board

Multiple choice paper (June 2022, Ordinary Level, 0570 Mathematics 1, one and a half hours). Answer ALL 50 questions; all questions carry equal marks. Each question has four suggested answers A, B, C and D; mark only one answer for each question. Non-programmable calculators are allowed.

Multiple choice questions - choose the one correct answer, A, B, C or D.

1. Simplifying $\frac{1}{3}$ of $(-3+12)$ gives

a) -5
b) 3
c) 5
d) -3

2. The arithmetic property for which $5+3=3+5$ is

a) Associative
b) Distributive
c) Commutative
d) Additive

3. The value of 4 in the number 37.49 is

a) 4 tenth
b) 4 tens
c) 4 hundreds
d) 4 units

4. The H.C.F. of $2a^2b$ and ab is

a) a^2b
b) $2a^2b$
c) ab
d) $2a^3b^2$

5. The number 80270 correct to one significant figure is

a) 80300
b) 80200
c) 81000
d) 80000

6. Given that a dress is sold for 6000 FCFA and a profit of 20% made on it. The cost price in FCFA is

a) 5000
b) 4800
c) 7200
d) 12000

7. An article is sold at a 20% profit on the cost price. The ratio of the cost price to the selling price is

a) $5:6$
b) $6:5$
c) $4:5$
d) $5:4$

8. A map is drawn to a scale of $1:2000$. The actual distance in meters, represented by 2.5 cm on the map is

a) 5000
b) 500
c) 5
d) 50

9. Expressing 0.32 as a fraction gives

a) $\frac{25}{8}$
b) $\frac{16}{5}$
c) $\frac{8}{25}$
d) $\frac{4}{125}$

10. Given that the exchange rate of 1 US dollar $= 550$ FCFA. Expressing 250 dollars in FCFA gives

a) 250000
b) 110000
c) 13750
d) 137500

11. Given the sets $M = \{a, b, c\}$ and $N = \{d, e\}$, then $M \cap N =$

a) 0
b) $\{\ \}$
c) $\{\emptyset\}$
d) $\{0\}$

12. A set $P = \{a, b\}$ is called a

a) Doubleton
b) A dual set
c) Power set
d) Proper set

13. Figure 1 is a Venn diagram, the universal set is

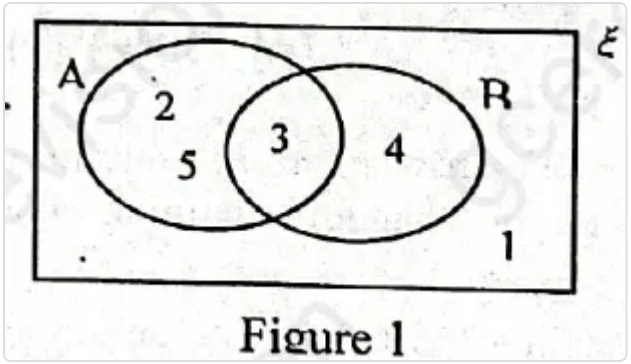


Figure 1

a) $\{2, 3, 4, 5\}$
b) $\{1\}$
c) $\{1, 2, 3, 4, 5\}$
d) $\{1, 3\}$

14. The converse of the notation $q \rightarrow p$ is

a) $\sim q \rightarrow \sim p$
b) $\sim p \rightarrow \sim q$
c) $p \rightarrow q$
d) $q \leftrightarrow p$

15. The mapping represented in figure 2 is

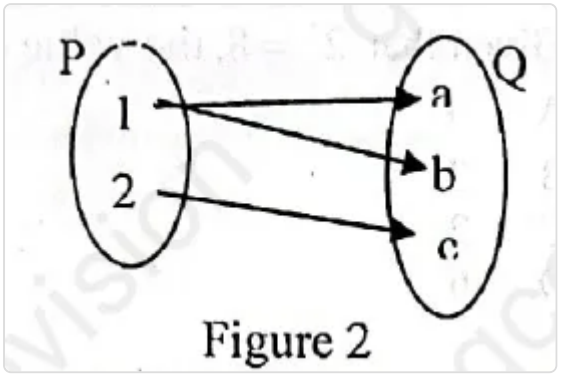


Figure 2

a) 1 - 1
b) 1 - Many
c) Many - 1
d) Many - Many

16. A function, f is defined on $\mathbb{R}$, a set of real numbers as $f : x \mapsto 2x - 1$. $f(-2) =$

a) 3
b) 5
c) -3
d) -5

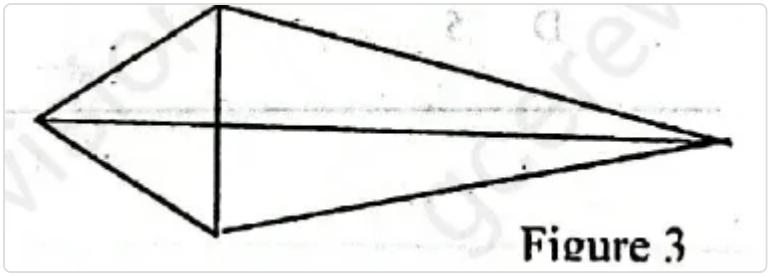
17. The inverse of the function $f : x \mapsto 2 + x$ is

a) $x - 2$
b) $2 - x$
c) $2 + x$
d) $-2 - x$

18. Given the lines l_1 and l_2 with equations $l_1 : 3x + 4 = y$ and $l_2 : y = 6x + 8$, then l_1 and l_2 are

Options as printed: A) Parallel B) Perpendicular C) Adjacent D) Transversals

19. The plane figure represented in figure 3 is a



a) Rhombus
b) Kite
c) Quadrilateral
d) Trapezium

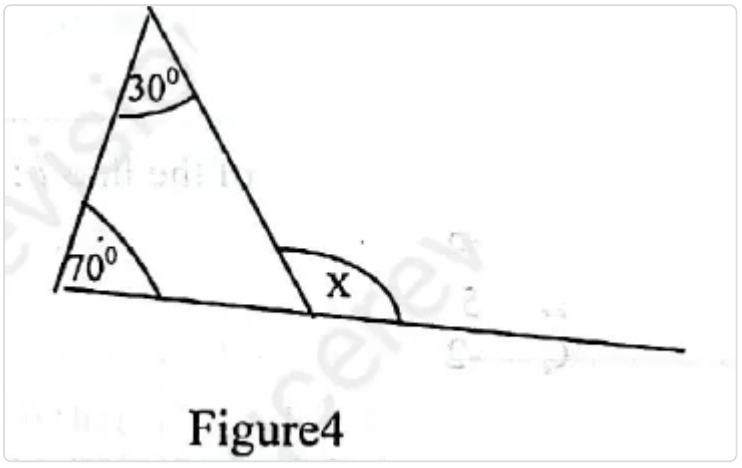
20. An interior angle of a regular plane figure is 90°, the number of sides is

a) 6
b) 3
c) 4
d) 5

21. A rectangle - based pyramid has

a) Eight faces
b) Four faces
c) Six edges
d) Five vertices

22. The angle x in figure 4 is



a) 100°
b) 80°
c) 30°
d) 70°

23. Figure 5 is a rectangular farm of area $1000\ \mathrm{m}^2$ and a width of $20\ \mathrm{m}$, the length, in m, is

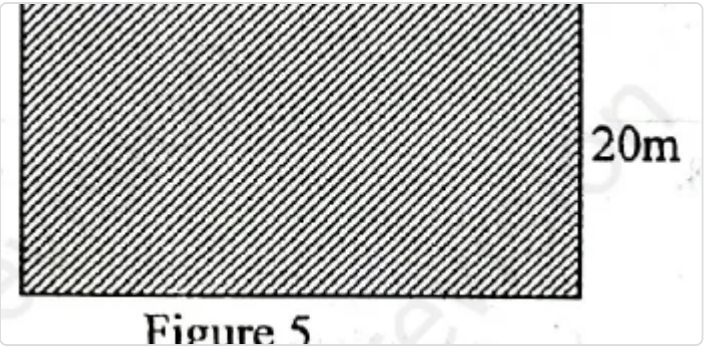


Figure 5

a) 80
b) 60
c) 50
d) 40

24. For any circle, the ratio $\frac{circumference}{diameter}$ is equal to

a) Radius (r)
b) Pi (π)
c) Diameter
d) Chord

25. Figure 6 is a cuboid of length 8 cm, width 5cm and height 4cm. The volume of the cuboid in cm^3 is

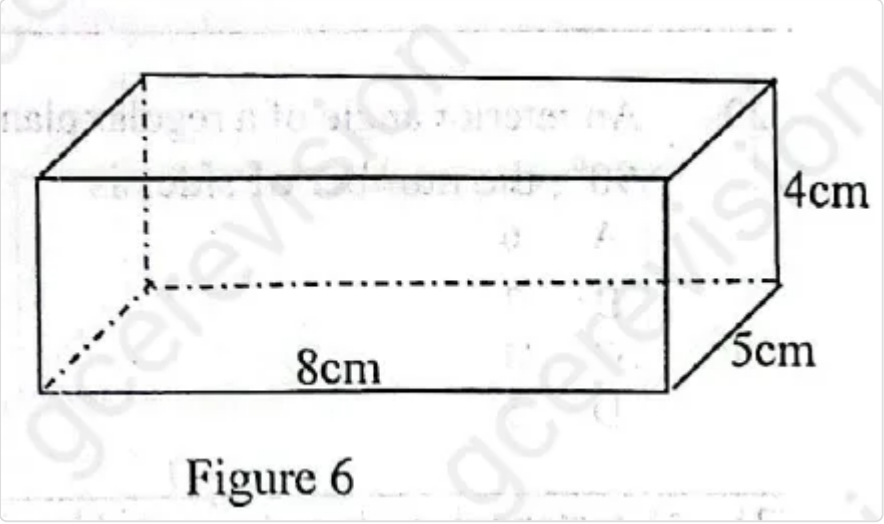


Figure 6

a) 17
b) 44
c) 72
d) 160

26. The x - intercept of the line $l : 2y + 5x = 10$ is

a) -5
b) 5
c) -2
d) 2

27. Given an equation of a line $2y + x = 3$. The gradient of the line is

a) $-\frac{1}{2}$
b) $\frac{1}{2}$
c) 2
d) -2

28. In the equation $1 + 4x = 9$, x is known as

a) A variable
b) An unknown
c) A solution
d) A constant

29. Simplifying $\frac{6m^2}{3m}$ gives

a) $2m^2$
b) $3m$
c) $2m$
d) $3m^2$

30. Expanding $(x + 3)(x + 3)$ gives

a) $x^2 + 3^2$
b) $x^2 + 3x + 9$
c) $x^2 + 9x + 9$
d) $x^2 + 6x + 9$

31. Given that $x = 3$ and $y = 2$, the value of $2x - 3y$ is

a) 12
b) 0
c) -2
d) -1

32. Given that $2^x = 8$, the value of x is

a) 4
b) 3
c) 2
d) 6

33. The inequality represented on the number line in figure 7 is

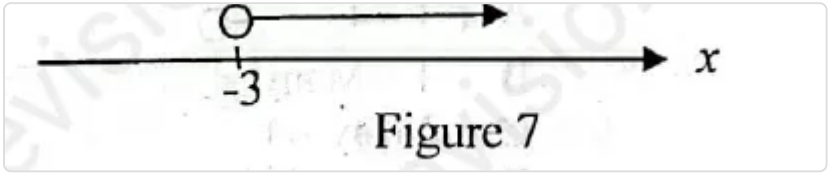


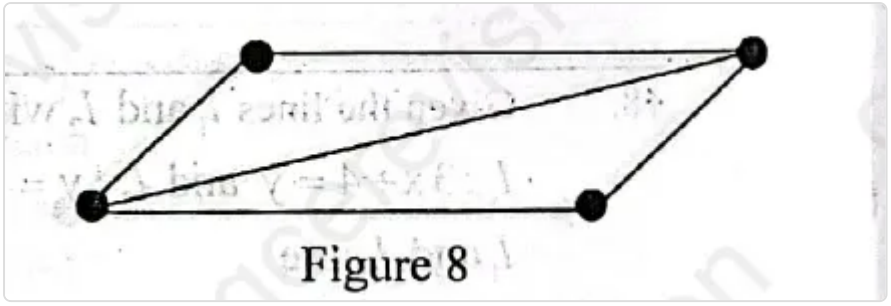
Figure 7

a) $x > -3$
b) $x \geq -3$
c) $x < -3$
d) $x \leq -3$

34. The nth term, T_n, of a sequence is $2n - 3$, then the 4th term is

a) -4
b) 4
c) 5
d) 11

35. Figure 8 is a network, the number of arcs is


Figure 8

a) 3
b) 2
c) 4
d) 5

36. Figure 9 is a quadratic graph of $y = f(x)$. The solution of $f(x) = 0$ is / are

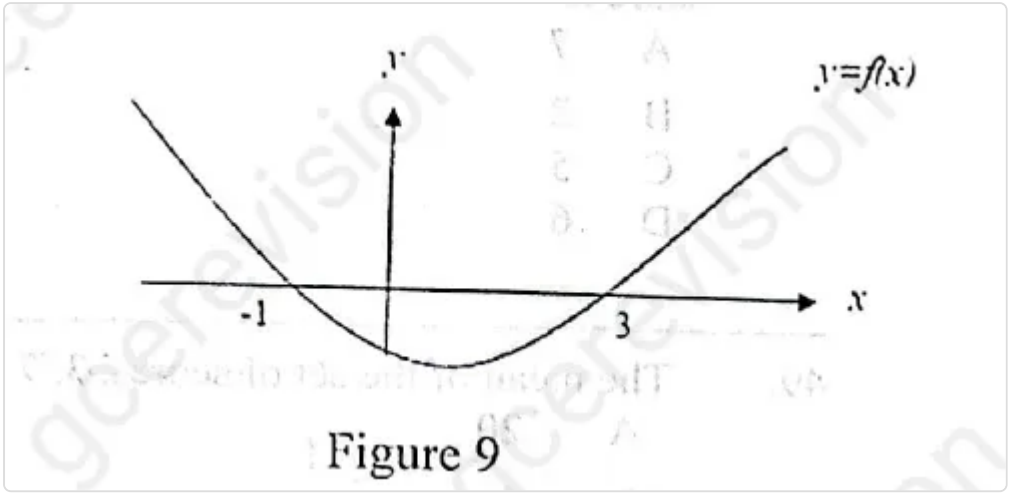


Figure 9

a) 3
b) -1 and 3
c) -1
d) 1 and -3

37. The trigonometric ratio which is positive in the fourth quadrant is

a) Cosine
b) Tangent
c) Sine
d) Cosecant

38. Given that figure 10 is a right - angled triangle, the value of p is

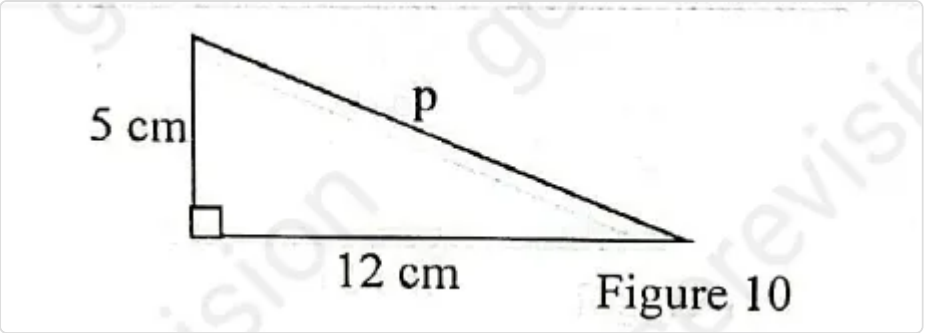


Figure 10

a) 60
b) 169
c) 13
d) 17

39. The value of $\tan 45^\circ$ is

a) $\frac{2}{\sqrt{3}}$
b) $\frac{1}{2}$
c) $\frac{1}{\sqrt{2}}$
d) 1

40. Given that $\vec{a} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, then $3\vec{a}$ is

a) $\begin{pmatrix} 6 \\ 9 \end{pmatrix}$
b) $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$
c) $\begin{pmatrix} 2 \\ 9 \end{pmatrix}$
d) $\begin{pmatrix} 5 \\ 6 \end{pmatrix}$

41. The magnitude of the vector $5j - 2j$ is

Options as printed: A) 3 B) $\sqrt{29}$ C) $\sqrt{21}$ D) 7

42. Given that $\overrightarrow{OA} = 3i + 4j$ and $\overrightarrow{OB} = 3i + 2j$, then $\overrightarrow{AB} =$

a) $6j + 6j$
b) $2j$
c) $-2j$
d) $3i + 3j$

43. A square matrix among the following is

a) $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$
b) $\begin{pmatrix} 2 & -3 & 4 \\ 5 & 0 & 1 \end{pmatrix}$
c) $\begin{pmatrix} a & b \end{pmatrix}$
d) $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

44. M is a 2 by 2 matrix such that $M = \begin{pmatrix} 0 & 5 \\ -2 & 1 \end{pmatrix}$, the determinant of M is

a) -10
b) 10
c) 3
d) -5

45. The image of the point $(3, 2)$ under the translation $\begin{pmatrix} -2 \\ 4 \end{pmatrix}$ is

a) $(1, 6)$
b) $(-6, 8)$
c) $(-5, 2)$
d) $(5, -2)$

46. The sum of all the angles of the sectors representing the items in any pie chart, is

a) 180°
b) 90°
c) 360°
d) 270°

47. The range of the set of scores, 3, 4, 1, 2, 5, 1, 8 is

a) 2
b) 7
c) 1
d) 8

48. Given the scores 2, 2, 7, 8 and 6. The median score is

a) 7
b) 2
c) 5
d) 6

49. The mean of the set of scores, 3, 7, 2 and 8 is

a) 20
b) 5
c) 4
d) 6

50. The probability of selecting a vowel from the word GOVERNANCE is

a) $\frac{2}{5}$
b) $\frac{3}{5}$
c) $\frac{3}{10}$
d) $\frac{7}{10}$

Want to check your answers?

The full correction of this paper, with the method explained step by step, is available on EduKora: 15,000 FCFA per child until the exam, 12,000 FCFA for each additional child in the same family.

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